

Introduction to Linear Regression

Why We Need Linear Regression

- Powerful and widely used statistical method.
- Models relationship between a dependent and one or more independent variables.
- Predicts continuous outcomes (e.g., house or stock prices).
- Simplifies understanding of variable relationships.
- Foundation for complex models in ML and statistics.

Theory of Linear Regression

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$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Y : Dependent variable
- X : Independent variable
- β_0 : Intercept
- β_1 : Slope
- ϵ : Error term

Assumptions of Linear Regression

Assumptions

- **Linearity**
- **Independence**
- **Homoscedasticity**
- **Normality**
- **No Multicollinearity** (for multiple regression)

Math Example: Linear Regression

Step 1: Given Data

We are given:

$$X = [1, 2, 3, 4], \quad Y = [5, 7, 9, 11]$$

We aim to find:

$$Y = \beta_0 + \beta_1 X$$

Step 2: Calculate the Means

$$\bar{X} = \frac{1 + 2 + 3 + 4}{4} = 2.5, \quad \bar{Y} = \frac{5 + 7 + 9 + 11}{4} = 8$$

Step 3: Compute Sample Covariance

$$\begin{aligned}\text{Cov}(X, Y) &= \frac{1}{3} [(1 - 2.5)(5 - 8) + (2 - 2.5)(7 - 8) + (3 - 2.5)(9 - 8) + \\ &\quad + (4 - 2.5)(5 - 8)] \\ &= \frac{1}{3}(4.5 + 0.5 + 0.5 + 4.5) = \frac{10}{3} \approx 3.33\end{aligned}$$

Step 4: Compute Sample Variance of X

$$\begin{aligned}\text{Var}(X) &= \frac{1}{3}[(1 - 2.5)^2 + (2 - 2.5)^2 + (3 - 2.5)^2 + (4 - 2.5)^2] \\ &= \frac{1}{3}(2.25 + 0.25 + 0.25 + 2.25) = \frac{5}{3} \approx 1.67\end{aligned}$$

Step 5: Compute Slope β_1

$$\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{3.33}{1.67} \approx 2$$

Interpretation: For each unit increase in X , Y increases by 2.

Step 6: Compute Intercept β_0

$$\beta_0 = \bar{Y} - \beta_1 \bar{X} = 8 - 2 \times 2.5 = 3$$

When $X = 0$, the predicted $Y = 3$.

Final Linear Equation

$$Y = 3 + 2X$$

This is the regression line.

Model Evaluation

- Mean Squared Error (MSE) measures how well a regression model predicts values.
- Formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Given Data

- **Data points**: $X = [1, 2, 3]$, $Y = [5, 7, 9]$
- **Slope** (β) = 2
- **Intercept** (α) = 3
- **Predicted Values**: $\hat{Y} = [5, 7, 9]$
- **Residuals**: $\epsilon = [0, 0, 0]$

Step-by-Step MSE Calculation

1. Compute Squared Errors:

$$(5 - 5)^2 = 0, \quad (7 - 7)^2 = 0, \quad (9 - 9)^2 = 0$$

2. Sum Up Squared Errors:

$$0 + 0 + 0 = 0$$

3. Compute MSE:

$$MSE = \frac{0}{3} = 0$$

Conclusion

- Since all predictions match actual values perfectly, MSE is ****zero****.
- No error in predictions, indicating a perfect fit.
- In real-world applications, a lower MSE suggests better model performance.

Summary

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- Models the relationship between X and Y
- Assumes linearity, independence, constant variance, normality
- Computation uses means, covariance, and variance
- Resulting model: $Y = 3 + 2X$

Multivariate Linear Regression

Introduction

Multivariable linear regression is a statistical technique used to model the relationship between two or more independent variables and a dependent variable. The goal is to find the best-fitting hyperplane that predicts the dependent variable based on the independent variables.

In this presentation, we will explore the steps involved in solving a multivariable regression problem with two independent variables X_1 and X_2 , and one dependent variable Y .

Regression Model

The multivariable linear regression model can be expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Where:

- Y is the dependent variable.
- X_1 and X_2 are the independent variables.
- β_0 is the intercept.
- β_1 and β_2 are the regression coefficients.
- ϵ is the error term.

Steps in Multivariable Regression

The process of solving multivariable regression consists of several steps:

1. **Calculate the means of the variables:**

$$\bar{X}_1, \bar{X}_2, \bar{Y}$$

2. **Compute the covariance and variance:**

$$\text{cov}(X_1, Y), \text{cov}(X_2, Y), \text{cov}(X_1, X_2)$$

3. **Calculate the regression coefficients:**

$$\beta_1 = \frac{\text{cov}(X_1, Y) \cdot \text{var}(X_2) - \text{cov}(X_2, Y) \cdot \text{cov}(X_1, X_2)}{\text{var}(X_1) \cdot \text{var}(X_2) - \text{cov}^2(X_1, X_2)}$$

$$\beta_2 = \frac{\text{cov}(X_2, Y) \cdot \text{var}(X_1) - \text{cov}(X_1, Y) \cdot \text{cov}(X_1, X_2)}{\text{var}(X_1) \cdot \text{var}(X_2) - \text{cov}^2(X_1, X_2)}$$

Example: Calculating Regression Coefficients

Given the following dataset:

X_1	X_2	Y
2	4	8
3	6	10
5	8	14

We calculate the mean values for X_1 , X_2 , and Y :

$$\bar{X}_1 \approx 3.33, \quad \bar{X}_2 = 6, \quad \bar{Y} \approx 10.67$$

Covariances and Variances

Next, we compute the covariances and variances:

$$\text{cov}(X_1, Y) = 4.665, \quad \text{cov}(X_2, Y) = 6, \quad \text{cov}(X_1, X_2) = 3$$

$$\text{var}(X_1) = 2.3333, \quad \text{var}(X_2) = 4$$

Calculate the Regression Coefficients

Now, we can calculate the regression coefficients β_1 and β_2 :

$$\beta_1 \approx \frac{4.665 \times 4 - 6 \times 3}{2.3333 \times 4 - 3^2} = 1.98$$

$$\beta_2 \approx \frac{6 \times 2.3333 - 4.665 \times 3}{2.3333 \times 4 - 3^2} = 0.01$$

The final regression equation is:

$$Y = \beta_0 + 1.98X_1 + 0.01X_2$$

Where β_0 can be calculated using:

$$\beta_0 = \bar{Y} - \beta_1\bar{X}_1 - \beta_2\bar{X}_2$$

Other Types of Linear Regression

1. Simple Linear Regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- One independent variable

2. Multiple Linear Regression

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

- Multiple predictors

3. Polynomial Regression

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_n X^n + \epsilon$$

- Models nonlinear trends

4. Ridge Regression (L2)

$$\text{Loss} = \sum (Y_i - \hat{Y}_i)^2 + \lambda \sum \beta_j^2$$

- Penalizes large coefficients

5. Lasso Regression (L1)

$$\text{Loss} = \sum (Y_i - \hat{Y}_i)^2 + \lambda \sum |\beta_j|$$

- Shrinks some coefficients to zero (feature selection)

6. Elastic Net Regression

$$\text{Loss} = \sum (Y_i - \hat{Y}_i)^2 + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

- Combines L1 and L2 regularization

7–9. Other Types

- Robust Regression
- Quantile Regression
- Stepwise Regression

Thank you!